

Table 3: Statistics of the 25 node-classification benchmarks used in the main experiment of this work.

Dataset	#Nodes	#Edges	#Features	#Classes	Feature Type	Train/Val/Test (%)	#Train	#Val	#Test	Homophily	Avg. Degree
Texas	183	741	1703	5	Binary Sparse	47.54/31.69/20.22	87	58	37	0.0609	4.05
Cornell	183	737	1703	5	Binary Sparse	47.54/32.24/20.22	87	59	37	0.1227	4.03
Wisconsin	251	1151	1703	5	Binary Sparse	47.81/31.87/20.32	120	80	51	0.1778	4.59
Chameleon	890	18598	2325	5	Binary Sparse	45.96/32.25/21.80	409	287	194	0.2361	20.9
Squirrel	2223	96219	2089	5	Binary Sparse	47.37/32.30/20.33	1053	718	452	0.2072	43.28
Wiki	2405	25597	4973	17	Unsigned Dense	14.14/42.91/42.95	340	1032	1033	0.6097	10.64
Cora	2708	13264	1433	7	Sparse	5.17/18.46/36.93	140	500	1000	0.8100	4.9
Citeseer	3327	12431	3703	6	Sparse	3.61/15.03/30.06	120	500	1000	0.7355	3.74
BlogCatalog	5196	348682	8189	6	Binary Sparse	2.31/48.85/48.85	120	2538	2538	0.4011	67.11
Actor	7600	60918	932	5	Binary Sparse	48.00/32.00/20.00	3648	2432	1520	0.2167	8.02
AmzPhoto	7650	245812	745	8	Binary Dense	2.09/48.95/48.95	160	3745	3745	0.8272	32.13
Minesweeper	10000	88804	7	2	One-hot	50.00/25.00/25.00	5000	2500	2500	0.6828	8.88
WikiCS	11701	442907	300	10	Signed Dense	4.96/15.12/49.97	580	1769	5847	0.6543	37.85
Tolokers	11758	1049758	10	2	Unsigned Dense	50.00/25.00/25.00	5879	2939	2940	0.5945	89.28
AmzComputer	13752	505474	767	10	Binary Dense	1.45/49.27/49.27	200	6776	6776	0.7772	36.76
DBLP	17716	123450	1639	4	Binary Sparse	0.45/49.77/49.77	80	8818	8818	0.8279	6.97
CoCS	18333	182121	6805	15	Sparse	1.64/49.18/49.18	300	9016	9017	0.8081	9.93
Pubmed	19717	108365	500	3	Binary Dense	0.30/2.54/5.07	60	500	1000	0.8024	5.5
FullCora	19793	146635	8710	70	Signed Dense	7.07/46.46/46.47	1400	9196	9197	0.5670	7.41
Roman Empire	22662	88516	300	18	Signed Dense	50.00/25.00/25.00	11331	5665	5666	0.0469	3.91
Amazon Ratings	24492	210592	300	5	Signed Dense	50.00/25.00/25.00	12246	6123	6123	0.3804	8.6
Deezer	28281	213785	128	2	Unsigned Dense	0.14/49.93/49.93	40	14120	14121	0.5251	7.56
CoPhysics	34493	530417	8415	5	Sparse	0.29/49.85/49.86	100	17196	17197	0.9314	15.38
Questions	48921	356001	301	2	Unsigned Dense	50.00/25.00/25.00	24460	12230	12231	0.8396	7.28
OGBN-Arxiv	169343	2484941	128	40	Signed Dense	53.70/17.60/28.70	90941	29799	48603	0.6542	14.67